## INTRODUCTION

We have come across many materials in our life, and they can be broadly classified into conductors, semiconductors and insulators based on their electrical conductivity. You might have noticed that conducting materials such as copper, aluminium etc., are used to transport electrical energy from one place to another place, and the insulating materials such as PVC, Bakelite etc., in switches, circuit boards etc., Do you know how the electrical energy is generated? We know from first Law of thermodynamics that energy can neither be created nor be destroyed, but one form of energy can be converted into another form. It is not possible to create electrical energy but we can generate electrical energy in many ways i.e., by converting solar energy, wind energy, tidal energy etc.... one such a way is converting chemical energy into electrical energy as in the case of batteries. We cannot imagine a modern technological world without batteries. Hence it is important to know the principles behind this type of energy conversion. The branch of chemistry that deals with the study of electrical energy transport and the inter conversion of electrical and chemical energy is called electrochemistry. Electrochemical reactions are redox reactions and they involve the transfer of electron from one substance to another.

In this unit, we will learn about the electrical conduction, construction of batteries and the thermodynamic principles involved in electro chemical reactions.

## 9.1 Conductivity of electrolytic solution

We have already learnt that when an electrolyte such as sodium chloride, potassium chloride etc... is dissolved in a solvent like water, the electrolyte is completely dissociated to give its constituent ions (namely cations and anions). When an electric field is applied to such an electrolytic solution, the ions present in the solution carry charge from one electrode to another electrode and thereby they conduct electricity. The conductivity of the electrolytic solution is measured using a conductivity cell. (Fig 9.1)

A conductivity cell consists of two electrodes immersed in an electrolytic solution. It obeys Ohm's law like metallic conductor. i.e., at a constant temperature, the current flowing through the cell (I) is directly proportional to the voltage across the cell (V).

$$\text{i.e., } I \propto V \text{ (or) } I = \frac{V}{R} \Rightarrow V = IR \qquad \text{.....(9.1)}$$

Where 'R' is the resistance of the solution in ohm (Ω)

Here the resistance is the opposition that a cell offers to the flow of electric current through it.

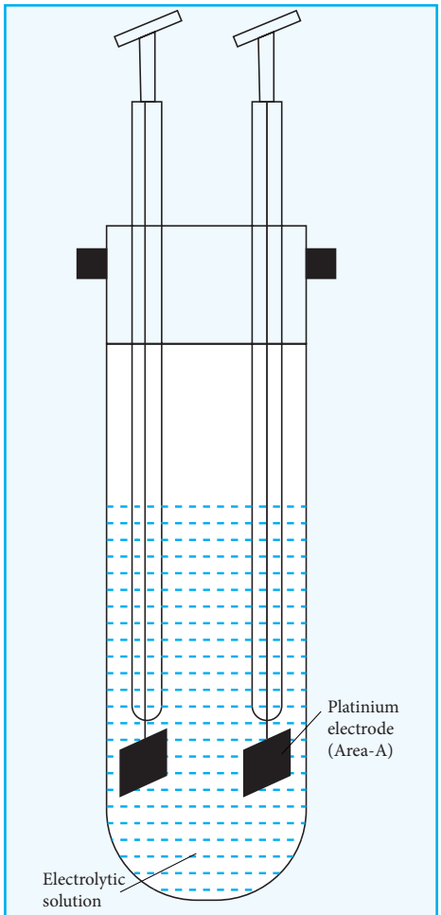


**Figure 9.1 conductivity cell**



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